STPM 2018 PHYSICS 960/3

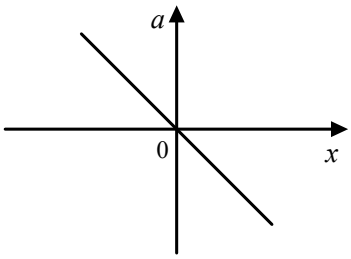
Questions and Answers

Values of constants: $h = 6.63 \times 10^{-34}$ J s, $c = 3.00 \times 10^{8}$ m s^{-1}, $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, 1 u $= 1.66 \times 10^{-27}$ kg.

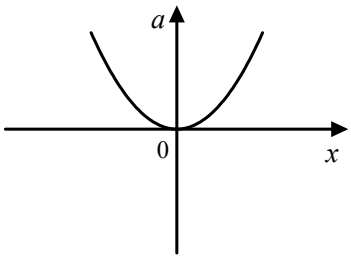
Section A [15 marks]

1. A particle performs a simple harmonic motion along the x-axis. Which graph shows the variation of acceleration, a, of the particle with displacement, x, from the equilibrium position?

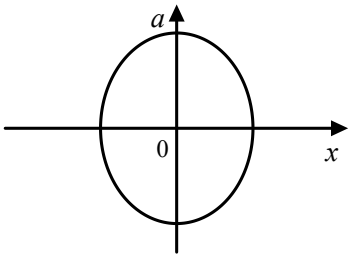
A



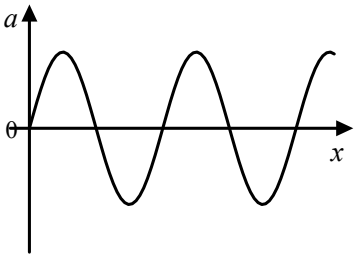
B



C



D



Answer: A

$a = -\omega^2 x$: a straight line through the origin with a negative gradient (A)

Explanation: The acceleration is always opposite to the displacement and proportional to it.

2. An external force is applied to a damped oscillatory system so that the system could oscillate continuously. Hence, to maintain the maximum amplitude of oscillation, the frequency of the applied force is

A half of the natural frequency of the oscillatory system
B double the natural frequency of the oscillatory system
C equal to the natural frequency of the oscillatory system
D independent of the natural frequency of the oscillatory system

Answer: C

The amplitude is greatest at resonance, when the driving frequency equals the natural frequency of the system.

Explanation: With damping, the resonance peak is slightly below the natural frequency, but to a good approximation they are equal.

3. The intensity of wave at a point, 2.0 m from the point source is 8.8×10^{-9} W m^{-2}. What is the intensity of wave at 3.5 m from the point source?

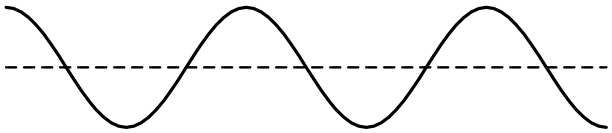
A 2.9×10^{-9} W m^{-2} B 5.0×10^{-9} W m^{-2} C 1.5×10^{-8} W m^{-2} D 2.7×10^{-8} W m^{-2}

Answer: A

$$I_2 = I_1(\frac{r_1}{r_2})^2 = (8.8 \times 10^{-9})(\frac{2.0}{3.5})^2 = 2.9 \times 10^{-9} \text{ W m}^{-2}$$

Explanation: Inverse square law for a point source.

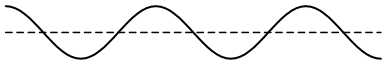
4. When a particle is at a maximum displacement, a stationary wave on a string is shown in the diagram below.



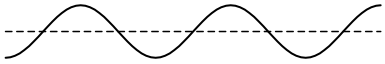
If the period of oscillation is T, which is the waveform of the stationary wave after $\frac{1}{8}T$?

A

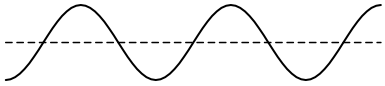
B



C



D



Answer: B

$y = 2A \cos kx \cos \omega t$: after $T/8$, $\cos(\pi/4) = 0.71$

The same shape (nodes in the same places) with 0.71 of the maximum displacement, on the same side: B.

Explanation: In a standing wave every point oscillates in phase (between nodes), so the shape keeps its nodes and simply shrinks towards the zero line.

5. The speed of wave in a stretched string depends on

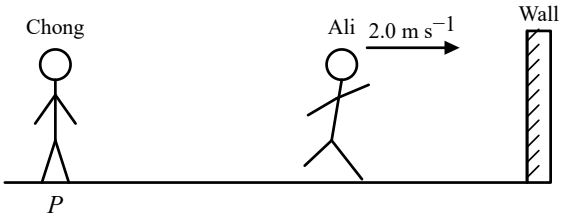
A the length and the tension of the string
B the amplitude of wave and the length of string
C the amplitude of wave and the tension of the string
D the tension and the mass per unit length of the string

Answer: D

$$v = \sqrt{\left(\frac{T}{\mu}\right)}\text{: it depends on the tension and the mass per unit length}$$

Explanation: The speed does not depend on the amplitude or the length of the string.

6. Ali and Chong stand stationarily at P. Ali then runs away from Chong with a speed of 2.0 m s^{-1} towards a vertical concrete wall while blowing a whistle that produces sound of frequency 500 Hz as shown in the diagram below.



The sound produced directly by Ali superposed with the sound reflected by the wall. What is the beat frequency heard by Chong?
[Speed of sound is 340 m s^{-1}.]

A 3 Hz B 5 Hz C 6 Hz D 9 Hz

Answer: C

$$\text{Direct (source receding): } f_1 = 500\frac{340}{340 + 2.0} = 497.1 \text{ Hz}$$

$$\text{Reflected (source approaching the wall): } f_2 = 500\frac{340}{340 - 2.0} = 503.0 \text{ Hz}$$

$$\text{Beat frequency} = 503.0 - 497.1 = 5.9 \approx 6 \text{ Hz}$$

Explanation: The wall reflects the frequency it receives, and Chong (stationary) hears it unchanged.

7. A shopkeeper observes a customer through a convex mirror with a radius of curvature, 20.0 m. What is the actual distance of the customer if the image of the customer is 4.5 m from the mirror?

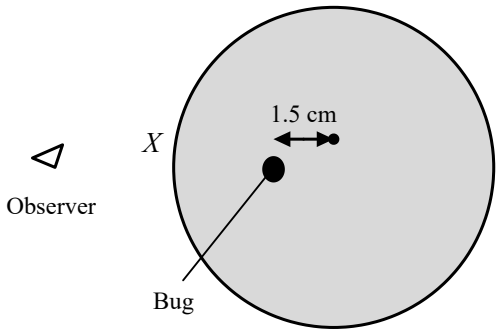
A 3.1 m B 3.7 m C 5.8 m D 8.2 m

Answer: D

$$f = -10.0 \text{ m}, v = -4.5 \text{ m}: \frac{1}{u} = \frac{1}{-10.0} - \frac{1}{-4.5} \Rightarrow u = 8.2 \text{ m}$$

Explanation: A convex mirror forms a virtual image behind it, closer than the object.

8. A sphere made of a material with a refractive index 1.5 contains a small bug 1.5 cm from the centre as shown in the diagram below.



Given that the radius of the sphere is 4.0 cm, what is the position of the image of the bug as seen by the observer from the left of surface X?

A 2.1 cm to the left of surface X
B 2.1 cm to the right of surface X
C 1.4 cm to the left of surface X
D 1.4 cm to the right of surface X

Answer: B

Bug is $4.0 - 1.5 = 2.5$ cm from surface X

$$\frac{1.5}{2.5} + \frac{1}{v} = \frac{1 - 1.5}{-4.0} \Longrightarrow v = -2.1 \text{ cm}$$

A virtual image 2.1 cm from X inside the sphere (to the right of X).

Explanation: Light goes from glass to air. Surface X is convex towards the observer, so its centre of curvature is on the incident (glass) side and $r = -4.0$ cm.

9. A plane-polarised light beams with intensity I_0 passes through a polaroid. The angle between the transmission axis of the polaroid and the plane of polarisation is θ. Which expression shows the correct intensity of the light, I, after passing through the polaroid?

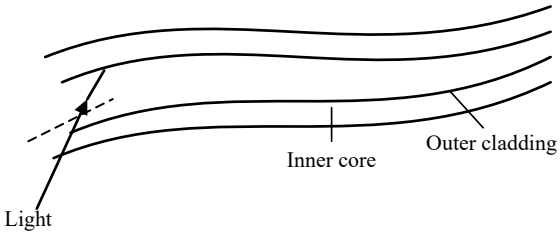
A $I = \frac{I_0}{2}\cos^2\theta$ B $I = I_0\cos^2\theta$ C $I = \frac{I_0^2}{4}\cos^2\theta$ D $I = I_0^2\cos^2\theta$

Answer: B

Malus's law: $I = I_0\cos^2\theta$

Explanation: The factor ½ applies only to unpolarised light passing through a polaroid.

10. A ray of light entering a fibre optics is shown in the diagram below.



Which statement is **not** true about the fibre optics above?

A The inner core is made of metal.
B The inner core is denser than the outer cladding.
C The angle of incidence is greater than the critical angle.
D The light ray in the inner core undergoes total internal reflection.

Answer: A

The core of an optical fibre is made of glass (or plastic), not metal. Statement A is not true.

Explanation: The core has a higher refractive index than the cladding, and light is guided by total internal reflection.

11. Photoelectrons emitted from a metal surface have a maximum kinetic energy which depends on

A the intensity of the incident radiation
B the amplitude of the incident radiation
C the wavelength of the incident radiation
D the area illuminated by the incident radiation

Answer: C

$K_{max} = \frac{hc}{\lambda} - W$: it depends on the wavelength (frequency) of the radiation

Explanation: Intensity, amplitude and illuminated area affect only the number of photoelectrons.

12. Which statement is **not** true about Bohr's atomic model for hydrogen atom?

A The electrons move about the nucleus in a circular orbit.
B The total energy of electron in the allowed orbit is constant.
C The angular momentum of electrons in the allowed orbit is quantized.
D The electrons radiate electromagnetic wave when they orbit the nucleus in the allowed orbit.

Answer: D

In Bohr's model, an electron in an allowed orbit does **not** radiate; it emits only when it jumps to a lower orbit. Statement D is not true.

Explanation: This postulate was Bohr's way round the classical prediction that orbiting electrons radiate.

13. The half-life of a radioactive isotope is 4.5 hours. If there are initially 2.00×10^{23} atoms of the isotope, the number of atoms of the isotope which decayed after 18 hours is

A 1.88×10^{23} B 1.75×10^{23} C 1.50×10^{23} D 2.50×10^{22}

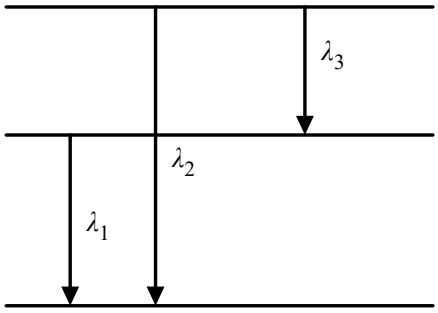
Answer: A

$$18 \text{ h} = 4 \text{ half-lives: remaining} = \frac{2.00 \times 10^{23}}{16} = 1.25 \times 10^{22}$$

$$\text{Decayed} = 2.00 \times 10^{23} - 0.125 \times 10^{23} = 1.88 \times 10^{23}$$

Explanation: The question asks for the number that has decayed, not the number remaining.

14. The transitions between energy levels for hydrogen atom produce three spectral lines of wavelengths, λ_1, λ_2 and λ_3, as shown in the diagram below.



If λ_1 and λ_2 are 122 nm and 103 nm respectively, what is the value of λ_3?

A 19 nm B 59 nm C 225 nm D 661 nm

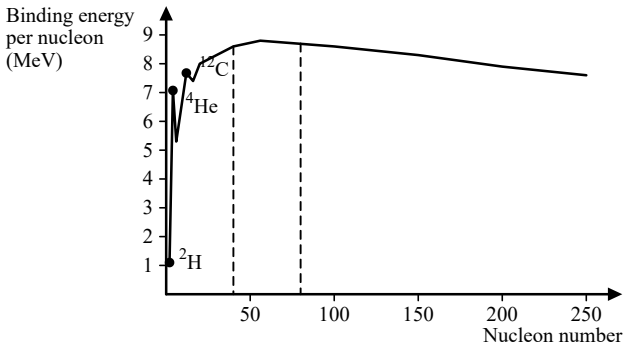
Answer: D

$$\frac{hc}{\lambda_2} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_3} \Rightarrow \frac{1}{\lambda_3} = \frac{1}{103} - \frac{1}{122}$$

$$\lambda_3 = 661 \text{ nm}$$

Explanation: Energies add (not wavelengths): the photon for the big jump equals the sum of the two smaller jumps.

15. The variation of binding energy per nucleon of the nucleus with nucleon number is shown in the graph below.



Which statement is true about the graph above?

A Nuclides with nucleon number between 40 and 80 are less stable.
B The lower the binding energy per nucleon, the more stable is the nucleus.
C Nuclides with high nucleon number will split to increase its binding energy per nucleon.
D Deuterium has the lowest binding energy per nucleon and undergoes fission to achieve its stability.

Answer: C

Heavy nuclides (high nucleon number) can split (fission) into nuclei with higher binding energy per nucleon, releasing energy.

Explanation: Nuclides with nucleon number 40–80 are the most stable; a higher binding energy per nucleon means more stable; deuterium undergoes fusion, not fission.

Section B [15 marks]

16. A firecracker explodes in the air at a height of 10.0 m vertically above a boy. The intensity level of the explosion heard by the boy is 60.0 dB.
[The threshold intensity of hearing is 1.00×10^{-12} W m^{-2}.]

(a) Calculate the power of sound for the explosion of the firecracker. [4]

Answer

$\beta = 10 \log \frac{I}{I_0} \Rightarrow 60.0 = 10 \log \frac{I}{1.00 \times 10^{-12}}$ 1

$I = 1.00 \times 10^{-6}$ W m^{-2} 1

$P = I(4\pi r^2) = (1.00 \times 10^{-6})(4\pi)(10.0)^2$ 1

$P = 1.26 \times 10^{-3}$ W 1

4

Explanation: Treat the explosion as a point source radiating equally in all directions.

(b) Determine the intensity level of the explosion heard by another boy which is 15.0 m from the explosion site. [1]

Answer

$I = \dfrac{1.26 \times 10^{-3}}{4\pi(15.0)^2} = 4.44 \times 10^{-7}$ W m^{-2}; $\beta = 56.5$ dB — 1

1

Explanation: Equivalently, 60.0 + 20 log (10.0/15.0) = 56.5 dB.

(c) Differentiate between the propagations of sound and light. [2]

Answer

Sound is a longitudinal mechanical wave that needs a medium; light is a transverse electromagnetic wave that can travel through a vacuum — 1

Sound travels much more slowly (about 340 m s^{-1} in air) than light (3.00×10^8 m s^{-1}) — 1

2

Total for question **7**

Explanation: This is why the flash is seen before the bang is heard.

17. A proposed controlled fusion process in a reactor is the Deuterium-Tritium fusion process represented by

$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}$.

(a) Determine the percentage of the combined mass of $^{2}_{1}$H and $^{3}_{1}$H which has been converted to energy. [2]

Answer

$\Delta m = (2.014102 + 3.016050) - (4.002625 + 1.008665) = 5.030152 - 5.011290 = 0.018862$ u — 1

Percentage $= \dfrac{0.018862}{5.030152} \times 100\% = 0.375\%$ — 1

2

Explanation: Only a very small fraction of the mass is converted, yet the energy released is enormous.

(b) Determine the energy released in the reaction. [2]

Answer

$E = (0.018862)(935 \text{ MeV}) = 17.6$ MeV — 1

$E = (17.6 \times 10^6)(1.60 \times 10^{-19}) = 2.82 \times 10^{-12}$ J — 1

2

Explanation: Use the conversion given in the question, 1 u ≡ 935 MeV.

(c) A city has an average power demand of 3.2×10^9 W. If the reactor has 35% efficiency, calculate the combined mass of ${}^2_1\text{H}$ and ${}^3_1\text{H}$ for 30 days.

[Mass of ${}^2_1\text{H}$ is 2.014102 u; mass of ${}^3_1\text{H}$ is 3.016050 u; mass of ${}^4_2\text{He}$ is 4.002625 u; mass of ${}^1_0\text{n}$ is 1.008665 u, 1 u ≡ 935 MeV.] [4]

Answer

Energy needed in 30 days = $(3.2 \times 10^9)(30 \times 24 \times 3600) / 0.35 = 2.37 \times 10^{16}$ J — 1

Number of reactions = $\dfrac{2.37 \times 10^{16}}{2.82 \times 10^{-12}} = 8.40 \times 10^{27}$ — 1

Mass of D + T per reaction = $(5.030152)(1.66 \times 10^{-27}) = 8.35 \times 10^{-27}$ kg — 1

Total mass = $(8.40 \times 10^{27})(8.35 \times 10^{-27}) = 70.1$ kg — 1

4

Total for question **8**

Explanation: With 35% efficiency, the reactor must release 1/0.35 times the electrical energy delivered.

Section C [30 marks]

18. (a) State three differences between progressive and standing waves. [3]

Answer

• A progressive wave transfers energy; a standing wave does not — 1

• In a progressive wave all particles have the same amplitude; in a standing wave the amplitude varies from zero (nodes) to a maximum (antinodes) — 1

• In a progressive wave the phase changes along the wave; in a standing wave all particles between adjacent nodes are in phase — 1

3

Explanation: Also accepted: the waveform of a progressive wave moves, while that of a standing wave does not.

(b) The equation of a progressive wave in a stretched string is given by $y = 0.020 \sin (4.0t - 5.0x)$, where x and y are measured in metre and t in second. Determine the speed of the wave. [4]

Answer

Compare with $y = A \sin (\omega t - kx)$: $\omega = 4.0$ rad s^{-1} — 1

$k = 5.0$ m^{-1} — 1

$v = \frac{\omega}{k}$ — 1

$v = 0.80$ m s^{-1} — 1

4

Explanation: Equivalently $f = \omega/2\pi$ and $\lambda = 2\pi/k$, then $v = f\lambda$.

(c) A standing wave is formed by the superposition of the progressive waves in (b) with the reflected wave from the ends where the string is tied. Assume there is no loss in energy. [Given that $\sin A + \sin B = 2 \cos ((A - B)/2) \sin ((A + B)/2)$.]

(i) State the principle of superposition. [1]

Answer

When two or more waves overlap, the resultant displacement at any point is the (vector) sum of the individual displacements — 1

1

Explanation: This holds at every instant and every point.

(ii) Deduce an equation for the standing wave formed. [3]

Answer

Reflected wave: $y_2 = 0.020 \sin (4.0t + 5.0x)$ — 1

$y = y_1 + y_2 = 2(0.020) \cos \frac{(4.0t - 5.0x) - (4.0t + 5.0x)}{2} \sin \frac{(4.0t - 5.0x) + (4.0t + 5.0x)}{2}$ — 1

$y = 0.040 \cos (5.0x) \sin (4.0t)$ m — 1

3

Explanation: $\cos (-5.0x) = \cos (5.0x)$. The position-dependent amplitude shows it is a standing wave.

(iii) Determine the amplitude of the standing wave at $x = 0.45$ m. [2]

Answer

Amplitude $= 0.040|\cos (5.0 \times 0.45)| = 0.040|\cos 2.25 \text{ rad}|$ — 1

$= 0.0251$ m — 1

2

Explanation: Use radians.

(iv) Determine the minimum length of the string used. [2]

Answer

The tied ends are nodes; the shortest string has one loop: $L = \frac{\lambda}{2} = \frac{\pi}{k}$ 1

$L = \frac{\pi}{5.0} = 0.628$ m 1

2

Total for question **15**

Explanation: $\lambda = 2\pi/k = 1.26$ m.

19. (a) State Huygens's principle. [2]

Answer

Every point on a wavefront acts as a source of secondary wavelets which spread out with the same speed as the wave 1

The new wavefront is the common tangent (envelope) to these wavelets 1

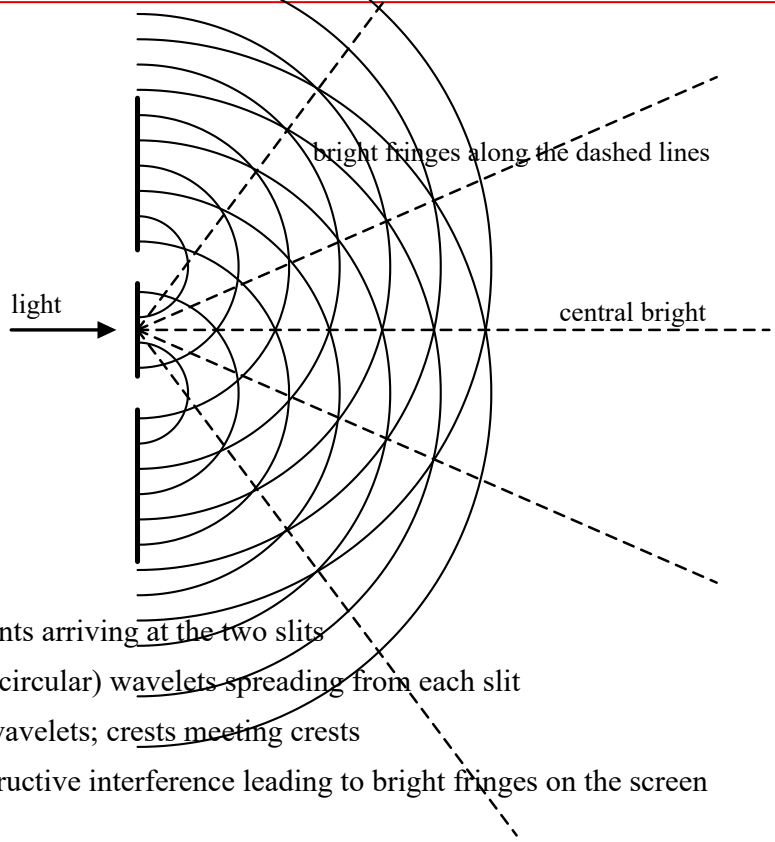
2

Explanation: Huygens's construction predicts how a wavefront moves forward.

(b) A monochromatic beam of wavelength 633 nm is incident perpendicularly onto a pair of slits. The separation of the slits is 12.0 μm. A series of bright and dark fringes are formed on the screen located at a distance of 2.8 m from the slits.

(i) Based on the Huygens's principle, sketch a diagram to show the formation of wave fronts which produce bright fringes on the screen. [4]

Answer



Plane wavefronts arriving at the two slits	1
Circular (semicircular) wavelets spreading from each slit	1
Overlapping wavelets; crests meeting crests	1
Lines of constructive interference leading to bright fringes on the screen	1
	4

Explanation: Bright fringes appear where the path difference from the two slits is a whole number of wavelengths.

(ii) Calculate the distance between two adjacent bright and dark fringes, on the screen. [3]

Answer

Fringe separation $y = \frac{\lambda D}{a} = \frac{(633 \times 10^{-9})(2.8)}{12.0 \times 10^{-6}} = 0.148$ m	1
Adjacent bright and dark fringes are half a fringe apart	1
Distance = 0.0739 m	1
	3

Explanation: With such a small slit separation the fringes are widely spaced.

(iii) What happen to the fringe width when a shorter wavelength monochromatic source is used? [1]

Answer

The fringe width decreases (fringes closer together)	1
	1

Explanation: $y \propto \lambda$.

(iv) Describe the change in fringe pattern if the beam from one slit has its phase shifted by half a wavelength relative to the beam from the other slit. [2]

Answer

The bright and dark fringes swap places: the centre becomes a dark fringe	1
The fringe separation is unchanged	1
	2

Explanation: An extra phase difference of π shifts the whole pattern by half a fringe.

(v) Explain the change in the interference pattern if the experiment is carried out in water. [3]

Answer

Light travels more slowly in water while its frequency is unchanged	1
so its wavelength becomes λ/n (about 0.75λ)	1
The fringe separation decreases by the same factor: the fringes move closer together	1
	3
Total for question	**15**

Explanation: The refractive index of water is about 1.33.

20. (a)

(i) Explain how continuous X-ray spectrum is produced. [3]

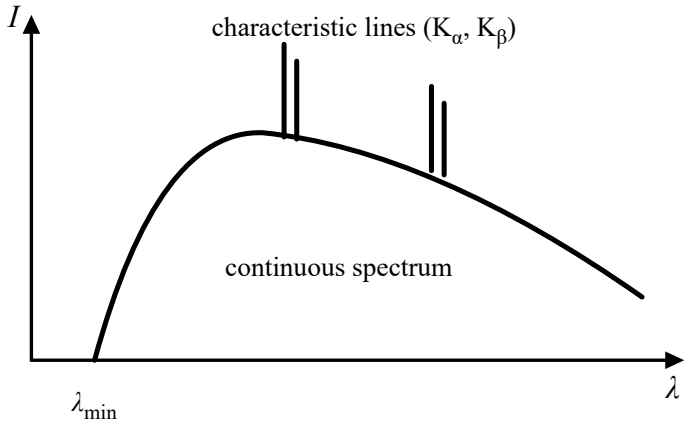
Answer

Electrons accelerated to high speed hit the metal target and are decelerated by the nuclei of the target atoms	1
The kinetic energy lost is emitted as X-ray photons (bremsstrahlung); different electrons lose different amounts, giving a continuous range of wavelengths	1
The minimum wavelength corresponds to an electron losing all its kinetic energy in one interaction: $eV = hc/\lambda_{min}$	1
	3

Explanation: Most of the electrons' energy becomes heat in the target.

(ii) Sketch and label a typical X-ray spectra. [2]

Answer



Continuous spectrum with a sharp cut-off λ_{min}	1
Characteristic lines superimposed and labelled	1
	2

Explanation: Both parts of the spectrum should be shown and labelled.

(b) In an X-ray tube, electrons are accelerated by a potential difference of 20.0 kV to hit a metal target.

(i) Determine the final speed of the electrons just before hitting the target. [2]

Answer

$\frac{1}{2}mv^2 = eV \Longrightarrow v = \sqrt{(\frac{2(1.60 \times 10^{-19})(20.0 \times 10^3)}{9.11 \times 10^{-31}})}$	1
$v = 8.38 \times 10^7$ m s^{-1}	1
	2

Explanation: Non-relativistic calculation (about 0.28c).

(ii) Calculate the minimum wavelength of the X-ray produced. [2]

Answer

$\lambda_{min} = \frac{hc}{eV} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{(1.60 \times 10^{-19})(20.0 \times 10^3)}$	1
$\lambda_{min} = 6.22 \times 10^{-11}$ m	1
	2

Explanation: All of the electron's kinetic energy becomes one photon.

(iii) Determine the de Broglie wavelength of the accelerated electrons. [2]

Answer

$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{(9.11 \times 10^{-31})(8.38 \times 10^{7})}$ 1

$\lambda = 8.68 \times 10^{-12}$ m 1

2

Explanation: The electron's de Broglie wavelength is much shorter than the X-ray wavelength it produces.

(c) The characteristic X-ray of wavelength 2.8×10^{-10} m is incident on a crystal. A first order diffraction is observed at a glancing angle of 12°. Determine the

(i) energy of the X-ray photon, [2]

Answer

$E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{2.8 \times 10^{-10}}$ 1

$E = 7.10 \times 10^{-16}$ J 1

2

Explanation: About 4.4 keV.

(ii) spacing between the atomic planes that caused the diffraction. [2]

Answer

$2d \sin\theta = \lambda \Longrightarrow d = \frac{2.8 \times 10^{-10}}{2 \sin 12°}$ 1

$d = 6.73 \times 10^{-10}$ m 1

2

Total for question **15**

Explanation: First order, $n = 1$; the glancing angle is measured from the planes.